

Electrospray ionization mass spectrometry (ESI-MS) employs a soft ionization technique that, unlike traditional ionization via electron bombardment, results in little to no fragmentation of target molecules. During electron bombardment, electrons collide with the analyte and cause the molecule to lose an electron whereas during ESI-MS, ions are produced by passing the analyte solution through an electrospray needle that has a potential difference applied with respect to a counterelectrode. This leads to the formation of charged droplets, which are repelled from the needle toward the counterelectrode and mass spectrometer, as shown in Figure 1.

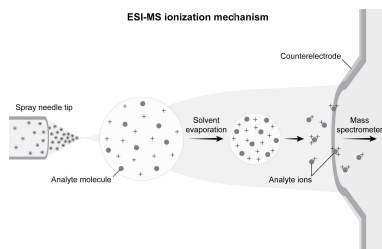


Figure 1 ESI-MS ionization mechanism

During the droplet's flight, the solvent is evaporated off analyte molecules. Ionization occurs through the protonation of basic sites on target molecules. Reaction 1 shows the number of protons n added to molecule M .



Reaction 1

Researchers used ESI-MS to quantify the formation of four pyridyloxobutyl-DNA (POB-DNA) adducts, shown in Figure 2. POB-DNA adduct formation involves highly mutagenic pyridyloxobutylating agents, which are formed from cytochrome P450-mediated hydroxylation of 4-(methylnitrosamino)-1-(3-pyridyl)-1-butanone (NNK), an abundant N-nitrosamine carcinogen found in cigarette smoke. Adducts were produced *in vitro* by treating samples of calf thymus DNA with NNK. The DNA adducts were then separated by reversed-phase high-performance liquid chromatography (RP-HPLC) and analyzed via ESI-MS.

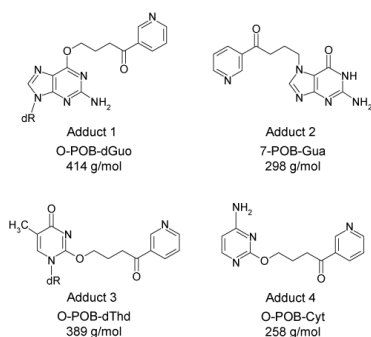


Figure 2 Molecular structure of POB-DNA adducts

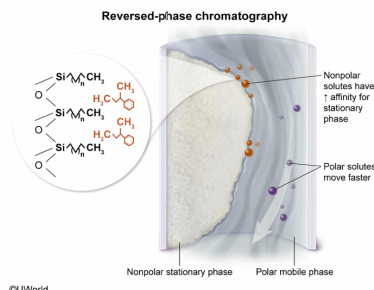
For the RP-HPLC separation of the POB-DNA adducts, which of the following would most likely be used as the mobile and stationary phases, respectively?

- ☐ A. Water and silica [10%]
- ☐ B. Hexane and C-18 alkyl hydrocarbon [8%]
- ☒ C. Water and C-18 alkyl hydrocarbon [53%]
- ☐ D. Hexane and silica [27%]

Correct

52% answered correctly

Explanation:



High-performance liquid chromatography (HPLC) is a purification technique in which a liquid solvent (mobile phase) is under pressure and carries a small sample through a column filled with adsorbent material (stationary phase) to a detector. As molecules pass through the

column, they are separated based on their relative **polarities** and the resulting differences in **interactions with the mobile and stationary phases**.

In **reversed-phase HPLC** (RP-HPLC), the **stationary phase** is a **nonpolar** compound, and the **mobile phase** is a comparatively **polar** solvent. Of the materials given, water (polar) would be appropriate for the mobile phase and C-18 alkyl hydrocarbon (nonpolar) would be appropriate for the stationary phase.

(Choices A and B) When the mobile and stationary phases are both polar (eg, water and silica) or both nonpolar (eg, hexane and C-18 alkyl hydrocarbon), analytes will not be separated effectively.

(Choice D) A nonpolar mobile phase (eg, hexane) and a comparatively polar stationary phase (eg, silica) are used in **normal-phase HPLC** (this type of HPLC is modeled after **thin-layer chromatography**). The passage and question state that *reverse-phase* HPLC was used to separate the POB-DNA adducts.

Educational objective:

High-performance liquid chromatography (HPLC) separates molecules according to their relative polarity and the resulting different interactions with the mobile and stationary phases. In reversed-phase HPLC, the mobile phase is a polar solvent and the stationary phase is made of a nonpolar material.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401566

System:
Separation Techniques, Spectroscopy, and Analytical Methods

Electrospray ionization mass spectrometry (ESI-MS) employs a soft ionization technique that, unlike traditional ionization via electron bombardment, results in little to no fragmentation of target molecules. During electron bombardment, electrons collide with the analyte and cause the molecule to lose an electron whereas during ESI-MS, ions are produced by passing the analyte solution through an electrospray needle that has a potential difference applied with respect to a counterelectrode. This leads to the formation of charged droplets, which are repelled from the needle toward the counterelectrode and mass spectrometer, as shown in Figure 1.

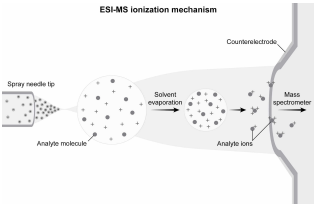
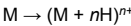


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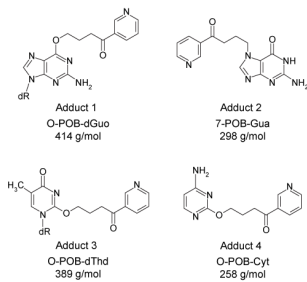
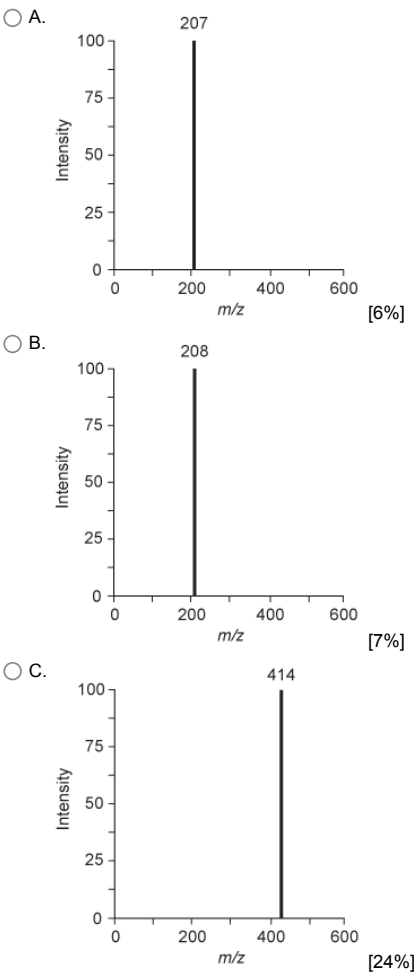
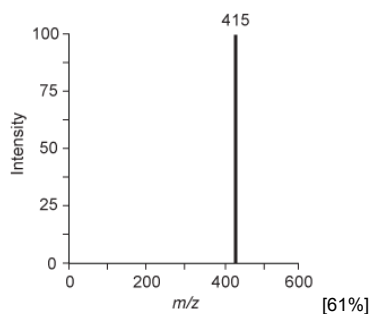


Figure 2 Molecular structure of POB-DNA adducts

Based on the procedure described in the passage, which of the following mass spectra most likely corresponds to the detection of the ion formed by the single protonation of POB-DNA Adduct 1?

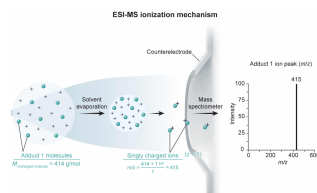


✓ ☒ D.



Correct
62% answered correctly

Explanation:

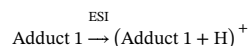


Mass spectrometry (MS) is a technique that **ionizes molecules**, detects the ion's molecular weight, and generates a plot (mass spectrum) of the **abundance** of each ion **versus** its **mass-to-charge ratio m/z** , which is calculated as

$$\frac{m}{z} = \frac{M_{\text{uncharged molecule}} + M_{\text{H}^+ \text{ added}}}{z}$$

where the numerator is the sum of the molecular weight of the uncharged molecule and the protons (H^+) added to the molecule, and z is the charge of the ion.

In the passage, MS is used to analyze ions produced by electrospray ionization (ESI). Figure 2 in the passage shows that the molecular weight of Adduct 1 is 414 g/mol, and the question specifies an ion formed by the **single protonation** of Adduct 1 during ESI.



Consequently, one proton has been added to the molecule (ie, the **mass** of the ion **increases by the mass of one H^+** , and **$z = +1$**). Using this information, the m/z ratio for the Adduct 1 ion can be calculated:

$$\frac{m}{z} = \frac{414 + 1}{1} = 415$$

Therefore, the mass spectrum for an ion formed by the **single protonation** of Adduct 1 ion has a peak with an m/z of 415. Choice D is the only spectrum that has a peak with an m/z of 415.

(Choice A) An m/z of 207 for Adduct 1 is obtained if it were doubly charged (ie, $z = +2$) and the mass of the two added protons was not taken into account (ie, $\frac{m}{z} = \frac{414}{2} = 207$).

(Choice B) A peak with an m/z of 208 occurs if Adduct 1 were doubly charged (ie, $z = +2$). The mass of the ion increases by the mass of two protons (ie, $\frac{m}{z} = \frac{414 + 2}{2} = 208$).

(Choice C) An m/z of 414 is obtained if only the mass of Adduct 1 is used and the mass of the added proton is not taken into account (ie, $\frac{m}{z} = \frac{414}{1} = 414$).

Educational objective:

Mass spectrometry is a technique that ionizes a molecule and detects the ions, generating a plot of the abundance of each ion vs. mass-to-charge ratio (m/z). For a singly charged protonated ion, a peak is observed with an m/z one mass unit greater than the molecular weight of the unprotonated molecule.

Time spent: 0 sec
Subject:
Organic Chemistry

QID: 401567
System:
Separation Techniques, Spectroscopy, and Analytical Methods

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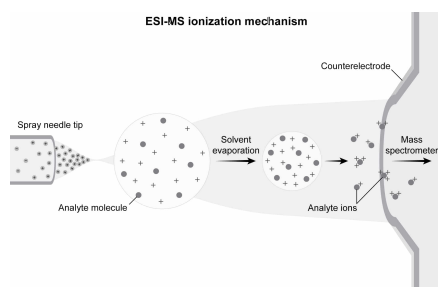


Figure 1 ESI-MS ionization mechanism

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Reaction 1

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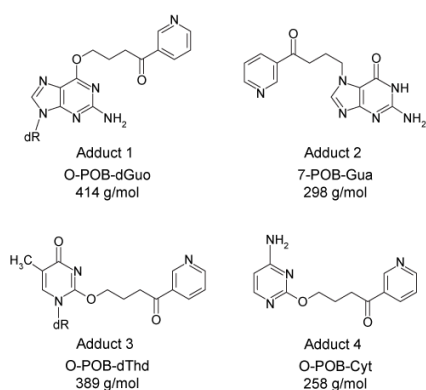


Figure 2 Molecular structure of POB-DNA adducts

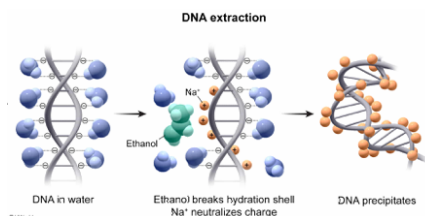
Prior to adduct formation, DNA was isolated from a cellular mixture. Membrane lipids were first extracted using an organic solvent, which was then evaporated, and DNA was precipitated from the aqueous layer using cold ethanol and sodium acetate. Which of the following mechanisms allowed for DNA precipitation?

- ☐ A. Sodium acetate lowers the pH of the solution, changing the conformation of the DNA structure. [7%]
- ☐ B. Ethanol has a greater polarity than water, pulling DNA out of the aqueous solution. [24%]
- ☐ C. Acetate ions interact favorably with the phosphate groups in DNA. [15%]
- ☒ D. Sodium cations form ionic bonds with DNA, neutralizing its charge. [52%]

Correct

52% answered correctly

Explanation:



Extraction is a technique that uses immiscible solvents (ie, an **organic solvent** and an **aqueous solution**) to separate molecules in a mixture based on **differences in solubility**. The principle "**like dissolves like**" indicates that molecules dissolve in a solvent with a similar polarity. **Hydrophilic molecules** containing several polar or ionic bonds have a greater affinity for the highly polar aqueous layer. Hydrophobic molecules such as lipids have fewer polar bonds, and therefore have a greater affinity for the less polar organic layer. A **molecule's charge or polarity** may be **modified** to **change** its **affinity** for the organic or aqueous layer.

The question states that DNA remains in aqueous solution after extraction of membrane lipids; this is due to negatively charged phosphate groups (PO_3^-) on the sugar phosphate backbone. The partially positive hydrogen atoms in water interact with the negatively charged DNA backbone through **ion-dipole interactions**, creating a hydration shell around the DNA.

To **precipitate the DNA** from aqueous solution, its **charge must be neutralized** through extraction with ethanol and a salt such as sodium acetate. Mixing the aqueous layer with ethanol disrupts the hydration shell around DNA molecules, allowing for sodium cations to better interact with the negatively charged DNA backbone. Sodium cations then neutralize DNA's charge via ionic bonding with phosphate groups, making DNA less hydrophilic, decreasing its affinity for the aqueous solvent, and allowing it to precipitate more efficiently.

(Choice A) **Sodium acetate**, the sodium salt of acetic acid, is a **weak base**. The addition of sodium acetate to water increases the solution's pH instead of reducing it.

(Choice B) Ethanol has a lower polarity than water. The addition of ethanol disrupts the solvation shell surrounding DNA, allowing for sodium cations to bond with DNA.

(Choice C) Both acetate and phosphate are negatively charged. Because **like charges repel** one another, acetate and phosphate will *not* interact favorably.

Educational objective:

Extraction is a technique that uses an organic solvent and an aqueous solution to separate molecules based on solubility. Molecules have a greater affinity for solvents with similar polarity ("like dissolves like"). Modifying a molecule's charge or polarity changes its affinity for the organic or aqueous layer.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401568

System:
Separation Techniques, Spectroscopy, and Analytical Methods

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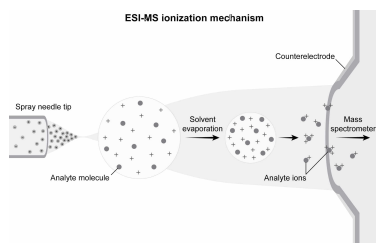
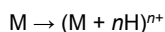


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Reaction 1

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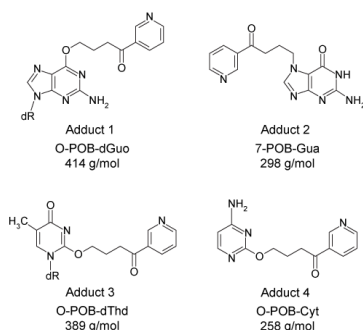


Figure 2 Molecular structure of POB-DNA adducts

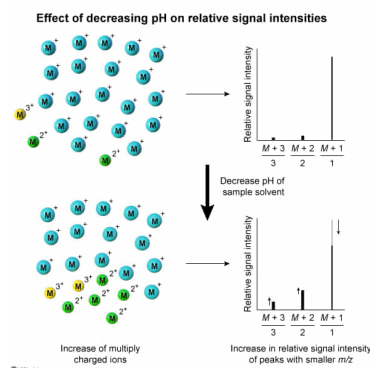
Which of the following experimental changes could be made to increase the relative signal intensity of the $(M + 2H)^{2+}$ and $(M + 3H)^{3+}$ ion peaks and decrease the relative intensity of the $(M + H)^{+}$ ion peak in the mass spectra of the POB-DNA molecules?

- ☐ A. Increase the pH of the sample solvent [23%]
- ☒ B. Decrease the pH of the sample solvent [60%]
- ☐ C. Decrease the concentration of the sample [6%]
- ☐ D. Increase the concentration of the sample [9%]

Correct

60% answered correctly

Explanation:



Mass spectrometry (MS) is a technique that **ionizes molecules**, detects the ion's molecular weight, and generates a plot (mass spectrum) of the **abundance** of each ion **versus** its **mass-to-charge ratio m/z** . In MS analysis, **signal intensity** (the height of a data peak relative to other peaks) corresponds to the relative **quantity of ions** at each mass-to-charge ratio m/z , which is calculated as

$$\frac{m}{z} = \frac{M_{\text{uncharged molecule}} + M_{\text{H}^+ \text{ added}}}{z}$$

where the numerator is the sum of the molecular weight M of the uncharged molecule and the protons (H^+ ions) added to the molecule, and z is the charge of the ion.

The passage states that during electrospray ionization, particles are charged via **protonation** within droplets. Molecular ions that have multiple charges (ie, are protonated multiple times) have a smaller m/z ratio because the denominator (charge) has increased. **Decreasing** the **pH** of the solvent increases the concentration of H^+ ions and results in a **greater number of multiply charged ions** (ie, a greater number of molecules are protonated multiple times to give $z > 1$). This decreases the number of molecules protonated once. Therefore, an increase in multiply charged molecular ions would result in greater signal intensity for smaller m/z values (eg, $\frac{m}{z} = \frac{M+2}{2}$ and $\frac{m}{z} = \frac{M+3}{3}$) and a decrease in the signal intensity for larger m/z values where $z = 1$.

(Choice A) An increase in the solvent's pH corresponds to a decrease in the concentration of H^+ ions, and therefore would not increase the number of multiply charged molecules or the signal intensity of the smaller m/z values.

(Choice C) Decreasing the concentration of the sample would decrease the amount of sample present and cause a decrease in the *overall* signal intensity.

(Choice D) Increasing the concentration of the sample would increase the amount of sample present, but this would not affect the relative heights of the peaks because all the peaks would increase.

Educational objective:

Signal intensity of a mass spectrometry peak corresponds to the relative quantity of ions at a given mass-to-charge ratio m/z . Increased charge yields a smaller m/z ratio; therefore, an increase in the number of multiply charged molecular ions increases signal intensity for smaller m/z ratios relative to the m/z peak where $z = 1$.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401569

System:
Separation Techniques, Spectroscopy, and Analytical Methods

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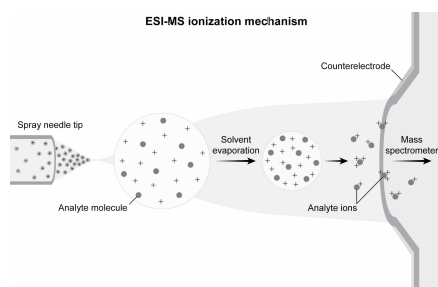


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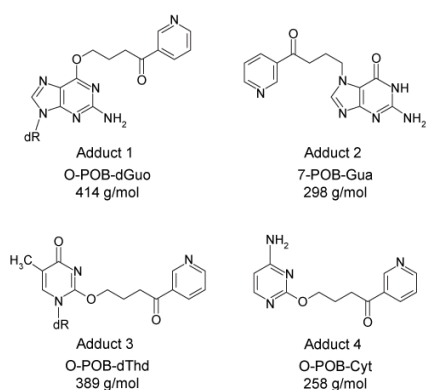
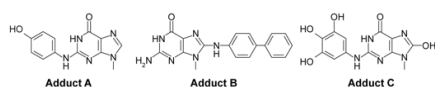


Figure 2 Molecular structure of POB-DNA adducts

In a second experiment, the following POB-DNA adducts are analyzed after separation by normal-phase HPLC. Which of the following is true regarding the separation of these adducts?

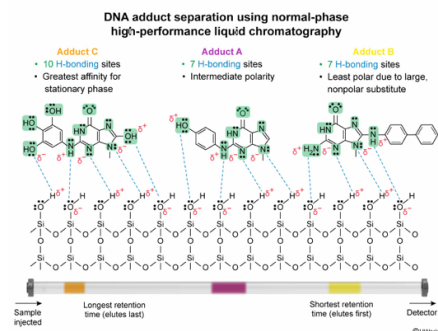


- ☐ A. Adduct A has a longer retention time than Adduct C. [4%]
- ☐ B. Adduct B has a longer retention time than Adduct C. [8%]
- ☒ C. Adduct C elutes last. [71%]
- ☐ D. Adduct B elutes last. [15%]

Correct

71% answered correctly

Explanation:



High-performance liquid chromatography (HPLC) is a purification technique that utilizes a stationary phase and a mobile phase of a substance. Molecules are separated based on their relative **polarities** and the resulting differences in the **interactions with the mobile and stationary phases**.

In **normal-phase HPLC (NP-HPLC)**, the **stationary phase** is **polar** relative to the **mobile phase**. Molecules with more polar bonds have a **stronger affinity** for the stationary phase and have longer retention times than molecules with fewer polar bonds, which have a stronger affinity for the relatively nonpolar (or less polar) mobile phase.

The question states that NP-HPLC is used to separate Adducts A, B, and C. These adducts each have atoms or groups of atoms that serve as **hydrogen bond donors and acceptors** and form interactions with the polar stationary phase:

- Each adduct has 4 hydrogen bond acceptors (1 O atom from a C=O bond and 3 N atoms).
- Each adduct also has groups of atoms that act as both a hydrogen bond donor and acceptor.
 - Adduct A: 3 groups (1 OH and 2 NH groups)
 - Adduct B: 3 groups (1 NH₂ and 2 NH groups)
 - Adduct C: 6 groups (4 OH and 2 NH groups)

Because Adduct C has a greater number of atoms capable of forming hydrogen bonds with the stationary phase, Adduct C has a greater affinity for the stationary phase than Adducts A and B. Therefore, Adduct C moves through the column slowest, has the longest retention time, and elutes last.

(Choices A and B) Adducts A and B each have *fewer* hydrogen bond donors to interact with the stationary phase than Adduct C. Therefore, Adducts A and B have *shorter* retention times than Adduct C.

(Choice D) Adduct B is the *least polar* adduct due to the large nonpolar **biphenyl substituent**, and would elute last if **reverse-phase HPLC** were used. However, the question states that NP-HPLC is used; therefore, Adduct B has the greatest affinity for the nonpolar mobile phase and elutes *first*.

Educational objective:

In normal-phase high-performance liquid chromatography, the stationary phase is polar relative to the mobile phase. Molecules with more polar bonds have a greater affinity for the stationary phase, and thus have a longer retention time than molecules with fewer polar bonds.

Time spent: 0 sec
Subject:
Organic Chemistry

QID: 401570
System:
Separation Techniques, Spectroscopy, and Analytical Methods